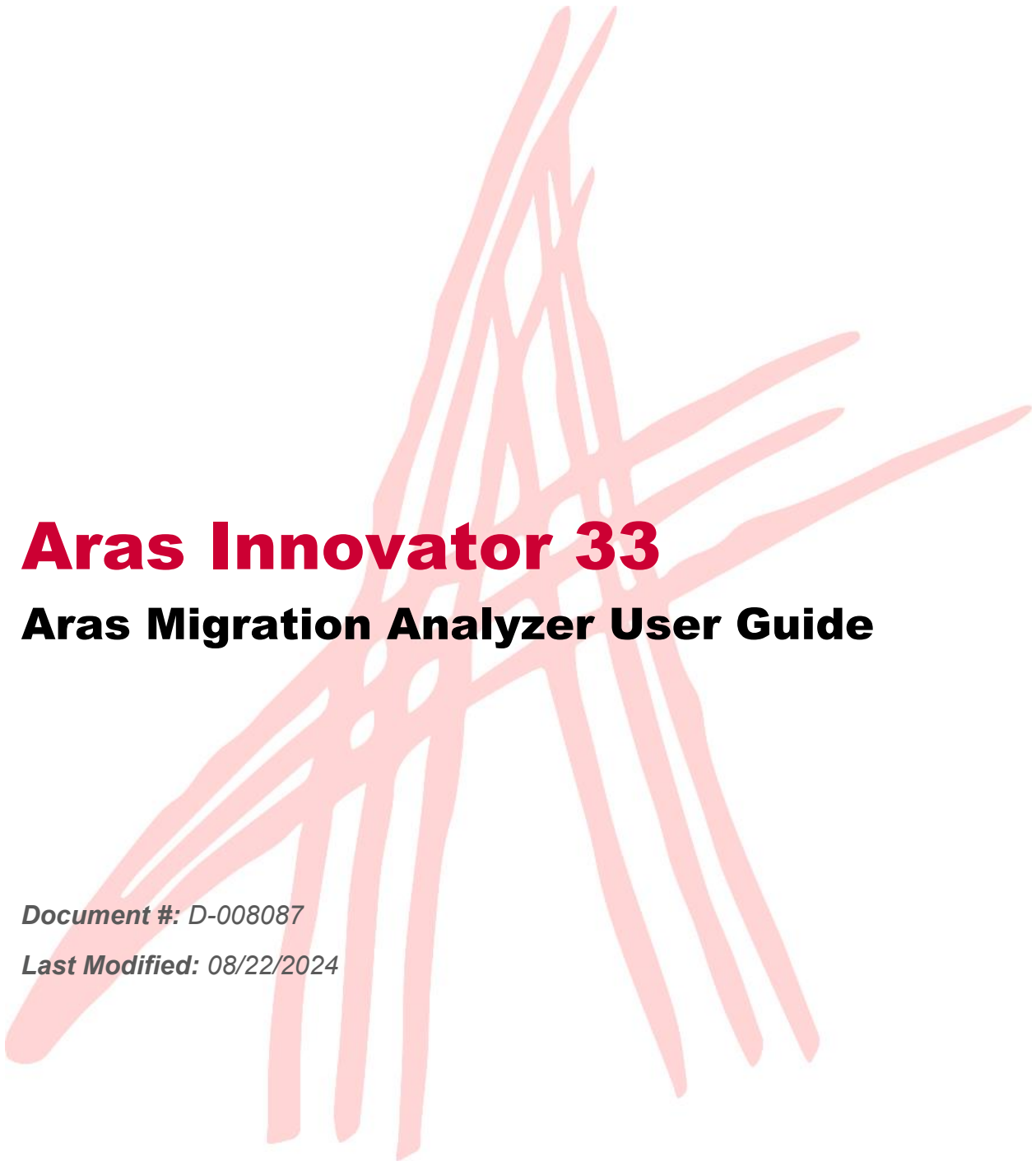




Aras Innovator 33

Aras Migration Analyzer User Guide

Document #: D-008087

Last Modified: 08/22/2024

Copyright Information

Copyright © 2024 Aras Corporation. All Rights Reserved.

Aras Corporation
100 Brickstone Square
Suite 100
Andover, MA 01810
Phone: 978-806-9400

E-mail: Support@aras.com

Website: <https://www.aras.com>

Notice of Rights

Copyright © 2024 by Aras Corporation and/or its affiliates. All rights reserved.

This document is protected by U.S. and international copyright laws and conventions. No copyright may be obscured or removed from this document. This document may not be modified or altered, or reproduced or transmitted in any form, without the explicit permission of the copyright holder.

Aras Innovator, Aras, and the Aras Corp "A" logo are registered trademarks of Aras Corporation in the United States and other countries.

All other trademarks referenced herein are the property of their respective owners.

Notice of Liability

THIS DOCUMENT IS PROVIDED FOR INFORMATIONAL PURPOSES ONLY, AND THE CONTENTS HEREOF ARE SUBJECT TO CHANGE WITHOUT NOTICE. THE INFORMATION CONTAINED IN THIS DOCUMENT IS DISTRIBUTED ON AN "AS IS" BASIS, WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE OR A WARRANTY OF NON-INFRINGEMENT. ARAS SHALL HAVE NO LIABILITY TO ANY PERSON OR ENTITY WITH RESPECT TO ANY LOSS OR DAMAGE CAUSED OR ALLEGED TO BE CAUSED DIRECTLY OR INDIRECTLY BY THE INFORMATION CONTAINED IN THIS DOCUMENT OR BY THE SOFTWARE OR HARDWARE PRODUCTS DESCRIBED HEREIN.

Table of Contents

Send Us Your Comments	4
1 Introduction	5
1.1 Purpose and Scope of This Guide	5
1.2 Target Audience	5
2 Prerequisite	5
3 How to Use the Aras Migration Analyzer	5
3.1 Setup	5
3.1.1 <i>Setting Up the Database</i>	5
3.1.2 <i>Setting Up the Code Tree</i>	6
3.2 Running the Method Code Analysis	6
3.3 Understanding the Analysis Results	10
3.3.1 <i>Reviewing Results in the Migration Analyzer</i>	10
3.3.2 <i>Reviewing Results in a PDF</i>	11
4 Troubleshooting	12
4.1 "Cannot find the AMA_UTILITYMethod"	12
4.2 "DebugServerMethod operating parameter not specified"	12

Send Us Your Comments

Aras Corporation welcomes your comments and suggestions on the quality and usefulness of this document. Your input is an important part of the information used for future revisions.

- Did you find any errors?
- Is the information clearly presented?
- Do you need more information? If so, where and what level of detail?
- Are the examples correct? Do you need more examples?
- What features did you like most?

If you find any errors or have any other suggestions for improvement, indicate the document title, and the chapter, section, and page number (if available).

You can send comments to us in the following ways:

Email:

TechDocs@aras.com

Subject: Aras Product Documentation

Or,

Postal service:

Aras Corporation
100 Brickstone Square
Suite 100
Andover, MA 01810
Attention: Aras Technical Documentation

If you would like a reply, provide your name, email address, address, and telephone number.

If you have usage issues with the software, visit <https://www.aras.com/support/>

1 Introduction

This section explains the purpose and scope of the guide and describes the target audience.

1.1 Purpose and Scope of This Guide

This guide describes how to use the Aras Migration Analyzer to help developers identify which of their server-side methods cannot be automatically ported.NET Core. This is important for subscribers and partners to understand as they prepare to upgrade or deploy solutions to Release 20 or later.

In Release 15, Aras migrated the Innovator platform from .NET Framework to .NET Core. This change was necessary not only to ensure we're using current technology to build our products, but also so we can provide new functionality like support for Linux.

It is not in the scope of the Aras Migration Analyzer utility or this guide to automatically update method code or suggest fixes. For more information about updating your code from .NET Framework to .NET Core, please visit Microsoft's migration guide: <https://docs.microsoft.com/en-us/dotnet/core/porting/>

1.2 Target Audience

This guide is written for developers who are responsible for migrating .NET Framework code to .NET Core.

2 Prerequisite

The Aras Migration Analyzer requires the .NET 6.0 SDK to be installed to run the utility. The SDK installer can be downloaded from the Microsoft website (<https://dotnet.microsoft.com/en-us/download/dotnet/6.0>) if the user does not have it installed.

3 How to Use the Aras Migration Analyzer

3.1 Setup

3.1.1 Setting Up the Database

Perform the following steps for each Aras Innovator database you want to analyze.

1. Unzip the Aras Migration Analyzer package. You can do this on the server or a client machine.

2. Open the Nash utility page for your Innovator installation in a web browser:
<http://{server}/{webalias}/Client/scripts/Nash.aspx>

User: Password: Database:

Password Hash Algorithm: ☒ MD5 ☐ SHA256

TimeZone: Culture:

Action: Server:

XML:

```
<AML>
<Item type="Method" id="20A077F8E6E24809908340B944423111" action="add">
<comments />
<execution_allowed_to keyed_name="World"
type="Identity">A73B655731924CD0B027E4F4D5FCC0A9</execution_allowed_to>
<method_code>
<![CDATA[
        CCO.Response.Clear();

        bool.TryParse(this.getProperty("is_readiness_check", ""), out bool isReadinessCheck);

        string tempFolderParam = serverConnection.GetOperatingParameter("ServerMethodTempDir",
") .ToString();
        string debugEnabledParam = serverConnection.GetOperatingParameter("DebugServerMethod",
") .ToString();
    ]>
    </method_code>
</Item>
</AML>
```

Figure 1. Nash.aspx

3. Enter credentials for an administrative user like admin or root and click **Login**.
4. In the Aras Migration Analyzer package, open **InnovatorSideUtility/AMA_UtilityMethod.xml**.
5. Copy the contents of AMA_UtilityMethod.xml and paste into the **XML** field in the Nash form.
6. Click **Submit** to add the method to the Aras Innovator database you want to analyze.

Note: Executing this AML more than once will result in a “Method already exists” error message. If this happens, move on to the next section.

3.1.2 Setting Up the Code Tree

Perform the following steps for each Aras Innovator installation you want to analyze.

1. Open the **InnovatorServerConfig.xml** file in the installation directory of your Aras Innovator instance.
2. Add the following line to the file if it does not exist. If the DebugServerMethod operating parameter does exist, make sure it is set to “true”.

```
<operating_parameter key="DebugServerMethod" value="true" />
```

3. Save the InnovatorServerConfig.xml file.

Note: After you finish running the analysis, be sure to come back to this file and set the DebugServerMethod operating parameter to “false”. Leaving the parameter enabled for long periods of time cause many log files to accumulate on the server.

3.2 Running the Method Code Analysis

1. In the Aras Migration Analyzer package, find **MigrationAnalyzer.WpfUI.exe** and run it to start the Aras Migration Analyzer.

2. Enter the url of your Aras Innovator instance and click **Login**.

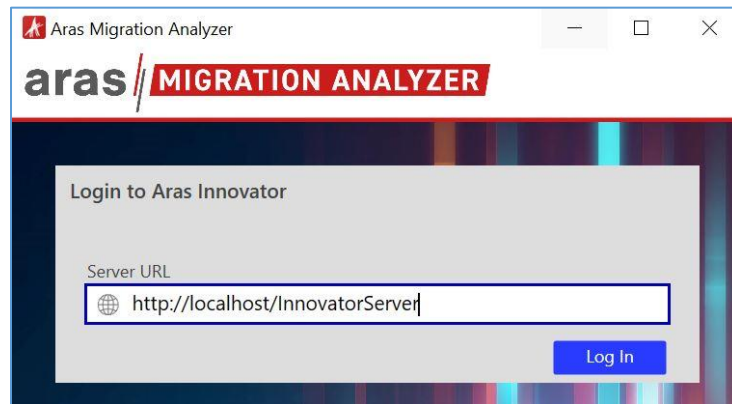


Figure 2. Aras Migration Analyzer login page

3. When the Aras Innovator login page appears, choose the database you want to analyze and enter credentials for an administrative user like admin or root. Click **Login**.

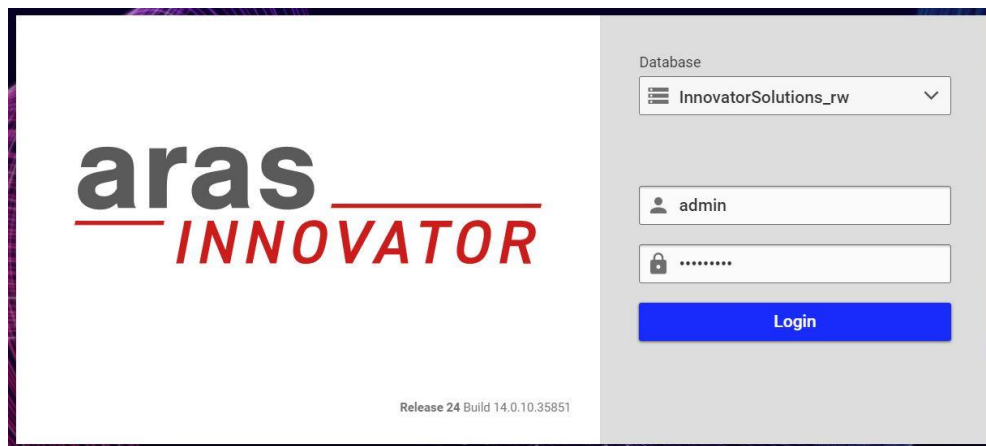


Figure 3. Aras Innovator login page

4. Next, the tool will present the main page for running the method code analysis.

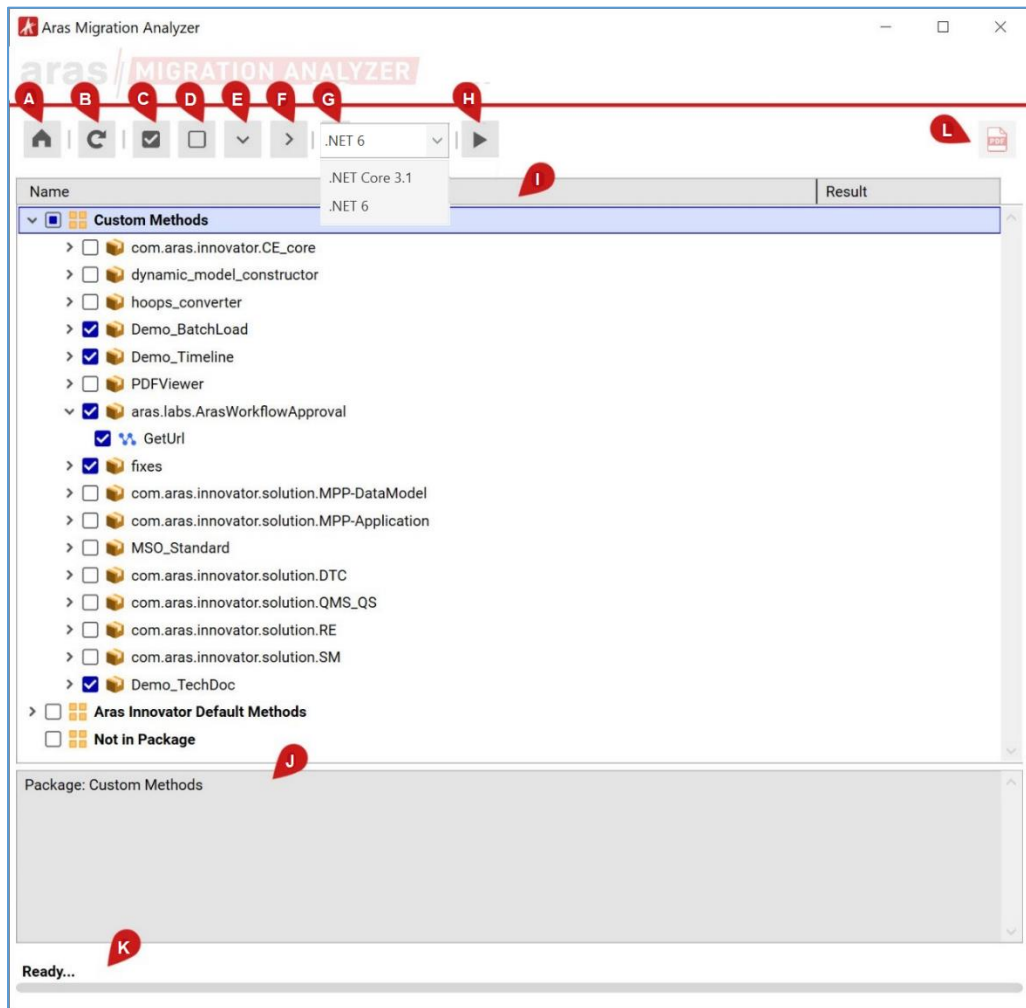


Figure 4. Main page for choosing methods to analyze

- a. **Home:** Return to the initial login page.
- b. **Refresh:** Reload the method list from the Aras Innovator database.
- c. **Select All:** Select every method in the list.
- d. **Deselect All:** Deselect every method in the list.
- e. **Expand All:** Expand all categories and packages to see every method in the list.
- f. **Collapse All:** Collapse all categories to hide all packages and methods in the list.
- g. **.Net Version:** Choose the .Net version (.NET 6 or .NET Core 3.1) to check the server methods for compatibility analysis.
- h. **Analyze:** Run the .NET Core migration analysis on the selected methods.
- i. **Method List:** This pane contains a list of server methods in the Aras Innovator database, organized by package. Packages are grouped into three categories:
 - i. **Custom Methods:** Contains all packages that are not core Aras Innovator packages. This includes Aras applications and custom packages.

- ii. **Aras Innovator Default Methods:** Contains core Aras Innovator packages. This category is deselected by default because Aras migrated all core code to .NET Core when subscribers upgraded to Release 15.
 - iii. **Not in Package:** Contains methods that don't belong to a package definition.
 - j. **Detail Pane:** Presents detailed analysis information about the selected method.
 - k. **Status Bar:** Displays the progress of the analysis. When the analysis is complete, the status bar indicates how many methods are compatible with .NET Core, incompatible, or have compilation errors.
 - l. **Export to PDF:** Allows the user to save the analysis results to a PDF file for review and sharing.
5. Select the methods you want to analyze from the method list and click the **Analyze** button.
 6. Wait until the status bar indicates that the analysis is complete. Now you're ready to review the results.

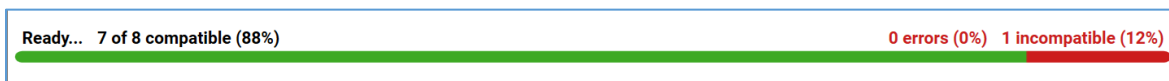


Figure 5. Status bar after analyzing 8 methods

3.3 Understanding the Analysis Results

There are two ways to review the results of the method analysis – inside the migration analyzer tool itself, or in an exported PDF.

Note: The Aras Migration Analyzer tool analyzes the compatibility of code only within the chosen server methods of an Aras Innovator database. It does not analyze code in custom DLLs that these server methods refer to. To assess the compatibility of code in a custom DLL, please use the relevant analyzer tool offered by Microsoft:

- The [API Port Utility](#) evaluates code for compatibility with .NET Core 3.1
- The [.NET Upgrade Assistant](#) evaluates code for compatibility against .NET 6.

3.3.1 Reviewing Results in the Migration Analyzer

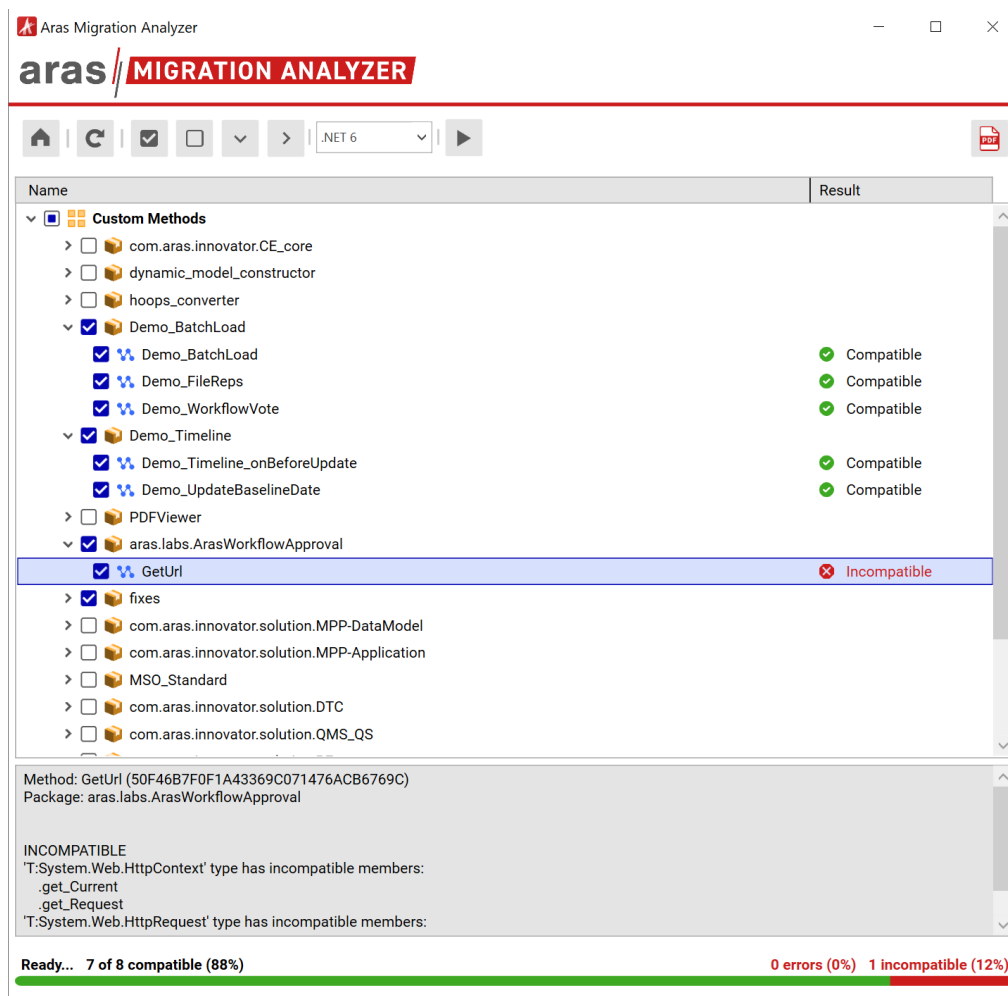


Figure 6. Completed method analysis in the Migration Analyzer UI

Once the analysis is complete, the righthand column of the method list will indicate one of the following results for each of the selected methods.

- **Compatible:** The code in this method is compatible with .NET Core and doesn't require updates.
- **Incompatible:** One or more .NET Framework APIs used in this method does not port to .NET Core. This code will need to be updated before it will work in Aras Innovator 15+. Click the method in the list and review the details pane to learn which specific APIs will need to be replaced.
- **Error:** There was a compilation error during the analysis. Click the method in the list and review the details pane for more information about the error.

For more information about migrating .NET Framework code to .NET Core, visit Microsoft's migration guide: <https://docs.microsoft.com/en-us/dotnet/core/porting/>

3.3.2 Reviewing Results in a PDF

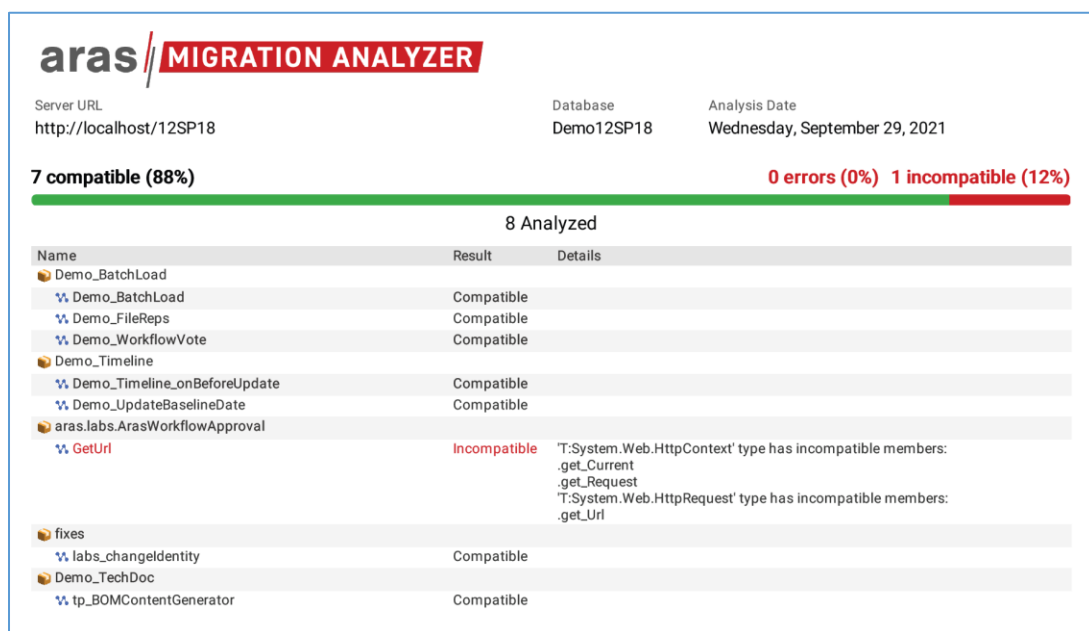


Figure 7. Completed analysis saved to PDF

Once the analysis is complete, you have the option to save the results to a PDF file. The file contains all the same information as the Migration Analyzer UI, but in an easily shared format. The PDF shows a result and details (where necessary) for each analyzed method.

4 Troubleshooting

4.1 “Cannot find the AMA_UTILITYMethod”

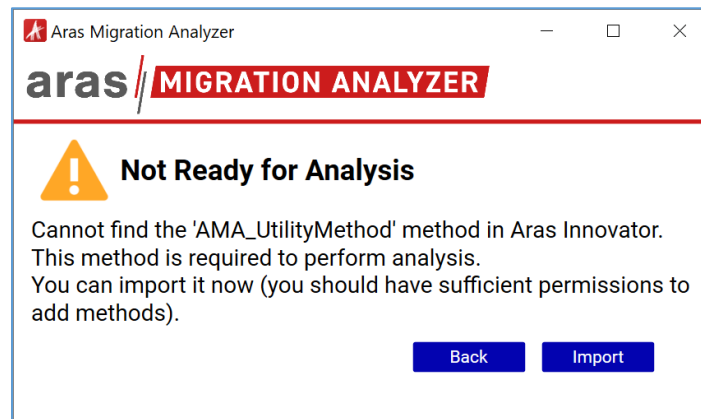


Figure 8. Error message about missing method

The Aras Migration Analyzer relies on the AMA_UTILITYMethod method to perform the analysis. This error message indicates that this method hasn't been added to the database you're trying to login to.

To resolve this error, either click the **Import** button shown in the screenshot above, or follow the manual steps detailed in section 2.1.1 Setting Up the Database.

4.2 “DebugServerMethod operating parameter not specified”

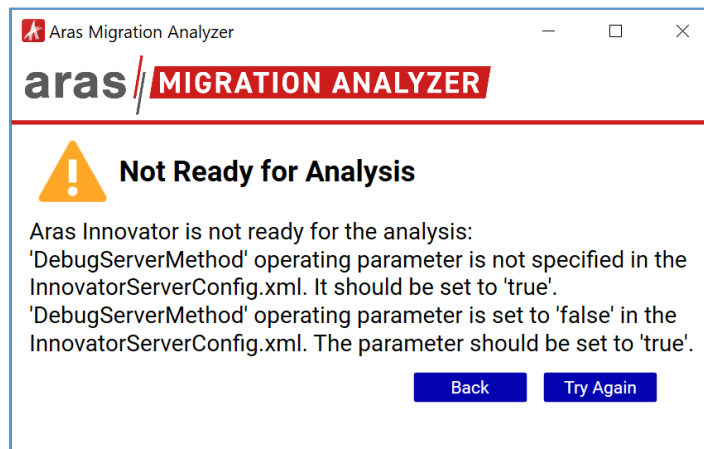


Figure 9. Error message about missing parameter

The Aras Migration Analyzer requires that the DebugServerMethod operating parameter is set to “true” in your instance's InnovatorServerConfig.xml file. This error message indicates that the parameter doesn't exist in that file, or it's not set to “true”.

To resolve this error, follow the steps detailed in section 2.1.2 Setting Up the Code Tree.